

11.SURGICAL MICROSCOPE

TECHNICAL SPECIFICATION

1. GENERAL REQUIREMENTS

Sl. No.	Parameter	Minimum Tender Requirement
1	Equipment	Advanced Surgical Operating Microscope
2	Applications	ENT, Otolaryngology, Neurotology, Skull Base, Neurosurgery, Spine and Microsurgery
3	Optical System	Premium apochromatic optical system
4	Visualization	High-resolution stereoscopic visualization
5	Main Surgeon	Binocular observation tube
6	Assistant	Co-observation facility
7	Magnification	Wide-range motorized magnification
8	Focus	Motorized/automated focus system
9	Working Distance	Minimum range covering 200–625 mm preferred
10	Illumination	Integrated high-intensity LED or Xenon illumination
11	Balancing	Automatic balancing system
12	Movement	Motorized microscope movement
13	Video	Integrated HD/4K video system
14	Display	Integrated touchscreen video monitor
15	Fluorescence	YELLOW 560 and INFRARED 800 options preferred
16	Navigation	Navigation integration capability
17	Networking	WLAN/Wi-Fi and LAN connectivity
18	Storage	Integrated internal storage, preferably 1 TB
19	PACS	DICOM/PACS connectivity
20	Control	Wireless foot control and hand controls
21	Sterility	Compatible with sterile draping system
22	Stability	Active vibration damping preferred
23	Warranty	Minimum 3 years comprehensive

2. OPTICAL SYSTEM

The microscope shall have:

- High-quality apochromatic optical system.

- High-resolution stereoscopic visualization.
- Excellent colour fidelity.
- High contrast.
- Excellent depth perception.
- Low optical distortion.
- Wide field of view.
- High light transmission.
- Stable visualization at high magnification.
- Long working distance suitable for ENT and neurosurgical applications.

Recommended requirement

- Working distance: **200–625 mm or better**
- 10× wide-field eyepieces.
- 12.5× wide-field eyepieces as preferred option.
- Foldable binocular tube **f170/f260 or equivalent**.
- 180° tiltable binocular tube or equivalent.
- Straight binocular tube option.

3. FOCUSING SYSTEM

The microscope shall have:

- Motorized focus.
- Automated focus adjustment.
- Rapid focus adjustment.
- Fine focus control.
- Focus control through handgrips.
- Foot control.
- Focus stability during prolonged procedures.

4. ILLUMINATION

The system shall provide:

- High-intensity coaxial illumination.
- Uniform illumination of the surgical field.
- Automatic illumination control.
- Long-life illumination source.
- LED or Xenon illumination.
- Automatic lamp exchange for Xenon configuration.
- Independent control of illumination intensity.
- Adequate illumination for deep surgical cavities.

5. AUTOMATIC BALANCING

The system shall have:

- Automatic balancing of the microscope.
- One-touch balancing function.
- Balance maintained after sterile draping.
- Easy repositioning.
- Stable microscope position.
- Automatic balance adjustment.

6. MICROSCOPE MOVEMENT

The system shall provide:

- Motorized X-Y movement.
- Motorized focus.
- Motorized magnification.
- Angular movement.
- Smooth positioning.
- Precise movement control.
- Joystick control.
- Foot pedal control.
- Electromagnetic brake system.
- Stable positioning during surgery.

Preferred: Active vibration damping for improved stability at high magnification.

7. ENT APPLICATIONS

The microscope shall be suitable for:

- Tympanoplasty.
- Mastoidectomy.
- Stapedectomy.
- Stapedotomy.
- Ossiculoplasty.
- Cochlear implant surgery.
- Middle-ear surgery.
- Neurotology.
- Skull-base surgery.
- Facial nerve procedures.
- Advanced otological microsurgery.

8. NEUROSURGICAL APPLICATIONS

The system shall be suitable for:

- Cranial microsurgery.
- Spinal microsurgery.
- Skull-base procedures.
- Brain tumour surgery.
- Vascular neurosurgery.
- Microsurgical decompression.
- Spine surgery.
- Other approved microsurgical procedures.

9. DIGITAL VIDEO SYSTEM

The system shall have:

- Fully integrated digital video camera.
- Minimum Full HD 1080p video.
- **4K UHD 2160p preferred.**
- Live video output.
- Still image capture.
- Video recording.
- Digital image enhancement.
- Image freeze.
- Brightness adjustment.
- Contrast adjustment.
- White balance.
- Digital zoom.
- External video output.
- Video-in facility.

For a premium tender configuration, specify:

Fully integrated 3-chip 4K camera with 2160p resolution or equivalent/superior technology.

10. INTEGRATED TOUCHSCREEN MONITOR

The system shall include:

- Integrated video touchscreen monitor.
- Monitor mounted on swivel arm.
- Multiple viewing angles.
- Easy viewing by surgeon and assistant.
- High-resolution display.
- Display of live surgical video.
- Control of microscope functions, where applicable.
- Touchscreen-based system control.

11. FLUORESCENCE IMAGING

For a high-end Government Hospital tender, the system should preferably include or be upgradeable to:

YELLOW 560

- Fluorescence-guided visualization.
- Compatible fluorescent agent as per regulatory approval.
- Rapid switching between white light and fluorescence.

INFRARED 800

- Near-infrared fluorescence imaging.
- ICG-based visualization.
- Vascular visualization.
- Intraoperative assessment.

The **Premium configuration** should preferably include:

- **YELLOW 560**
- **INFRARED 800**

Tender note: Fluorescence agents shall be used only according to the regulatory approval status applicable in India.

12. CO-OBSERVATION

The system shall provide:

- Main surgeon binocular viewing.
- Assistant surgeon co-observation.
- Face-to-face co-observation option.

- Left/right co-observer option.
- Stereo co-observation.
- High-quality stereoscopic image sharing.

The exact co-observation configuration shall be specified by the bidder.

13. NAVIGATION INTEGRATION

The microscope shall be capable of integration with:

- Surgical navigation systems.
- Cranial navigation.
- Spinal navigation.
- DICOM-compatible navigation systems.
- Third-party navigation systems, where supported.

The system shall provide an interface for navigation-guided procedures.

14. NETWORKING & DATA STORAGE

The system shall include:

- Integrated internal storage.
- Minimum **1 TB storage preferred**.
- WLAN connectivity.
- Wi-Fi hotspot capability.
- LAN connectivity.
- Shared network storage.
- Patient data management.
- Digital image storage.
- Video storage.
- DICOM/PACS integration.
- Network transfer.

15. FOOT CONTROL

The system shall include a **wireless multifunction foot control panel** capable of controlling, as applicable:

- Microscope movement.
- Focus.

- Magnification.
- Illumination.
- Image capture.
- Video recording.
- Fluorescence selection.
- Programmable microscope functions.

16. STERILE DRAPING

The system shall support:

- Sterile microscope draping.
- Easy draping.
- Secure sterile covers.
- Compatibility with Auto Drape or equivalent.
- Minimal movement restriction after draping.

17. SAFETY & QUALITY

The system shall comply with applicable:

- IEC 60601-1.
- IEC 60601-1-2.
- Applicable medical electrical equipment standards.
- Applicable Indian regulatory requirements.
- Manufacturer's quality and safety requirements.

18. REQUIRED ACCESSORIES

The complete system shall be supplied with:

1. Surgical Microscope – **1 No.**
2. Motorized floor stand – **1 No.**
3. Main surgeon binocular tube – **1 No.**
4. Assistant/co-observation tube – **1 No.**
5. 10× wide-field eyepieces – **1 pair**
6. 12.5× wide-field eyepieces – **1 pair**, preferred
7. Foldable f170/f260 tube – **1 No.**
8. Suitable objective lenses – **as required**
9. Motorized focus system – Integrated
10. Motorized magnification – Integrated
11. Auto Balance – Integrated
12. High-intensity illumination – Integrated

13. 3-chip 4K camera – **1 No.**
14. Integrated touchscreen monitor – **1 No.**
15. 4K 3D/2D visualization system, if offered
16. Wireless foot control – **1 No.**
17. Handgrip controls – **1 set**
18. YELLOW 560 fluorescence – **1 No.**, preferred
19. INFRARED 800 – **1 No.**, preferred
20. Navigation interface – **1 No.**
21. DICOM/PACS interface – **1 No.**
22. Integrated 1 TB storage – **1 No.**
23. WLAN/Wi-Fi connectivity – Included
24. Sterile microscope drapes – Initial supply
25. Dust cover – **1 No.**
26. User manual – **1 No.**
27. Service manual – **1 No.**
28. Installation and commissioning – Included
29. Application training – Included

19. WARRANTY

Parameter	Minimum Tender Requirement
Comprehensive Warranty	Minimum 3 years
Optical System	Covered
Microscope Stand	Covered
Camera	Covered
Illumination	Covered
Motorized Movement	Covered
Touchscreen Monitor	Covered
Fluorescence Modules	Covered
Video System	Covered
Service Response	Within 24–48 hours
Preventive Maintenance	Minimum 2 visits/year
Spare Parts	Minimum 7–10 years availability
Service Support	Authorized service support in India
Training	ENT surgeons, Neurosurgeons, OT staff & Biomedical Engineer
Installation	Complete installation and commissioning
Calibration	As per manufacturer's recommendation

- **Comprehensive Warranty:** Minimum 3 years from the date of installation/commissioning.

- **Preventive Maintenance:** Minimum 4 preventive maintenance visits per year during the warranty period.
- **Spare Parts Support:** Availability and support for genuine spare parts for a minimum period of 10 years.
- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.